

Hospital infection control data construction

Dataset construction explanation

This is a **synthetic teaching dataset**, not patient or hospital data. The two files are `hospital_wards.csv` (120 ward rows) and `hospital_transfers.csv` (420 directed transfer links). `generate_datasets.py` is the separate, staff-facing reproducibility script; it is never called by the student notebook. NumPy's `default_rng(2041)` fixes the random draw sequence.

Timeline and column definitions

The observation point is the end of a 14-day recording window. Transfers and the two ward attributes are available by that point. `active_cluster_at_start` is the cluster status at that point. `new_cluster_later` records whether a **new simulated cluster** is reported in the subsequent 14 days, only for wards without an active cluster at the observation point. Here a cluster is simply a synthetic binary label; no cases, pathogen or clinical threshold is modelled. Blank target values for wards already affected mean *ineligible*, not *no later cluster*.

File and column	Meaning
Wards: <code>ward_id</code>	Unique identifier W001 to W120; for joining, never a predictor.
Wards: <code>occupancy_ratio</code>	Simulated occupied-bed fraction, drawn uniformly from 0.48 to 0.96, rounded to three decimals.
Wards: <code>device_use_share</code>	Simulated fraction of patients with an invasive device, drawn uniformly from 0.07 to 0.48, rounded to three decimals.
Wards: <code>active_cluster_at_start</code>	Exactly 18 uniformly chosen wards have value 1 at the observation point.
Wards: <code>new_cluster_later</code>	Later binary outcome drawn only for the other 102 wards; blank for the 18 initially affected wards. There are 39 later positives in this fixed release.
Transfers: <code>source_ward_id</code> , <code>destination_ward_id</code>	Ordered sender and receiver for a pre-observation transfer connection.
Transfers: <code>transfer_count</code>	Synthetic number of transfers on that ordered link during the recording window, integer 1 to 9.

Graph and target rule

Sample 420 distinct ordered pairs uniformly without replacement from all pairs of different wards. For each pair, independently draw a transfer count of 1 to 9. Build a directed graph of all 120 wards; add the pair as a link with `transfer_count` as its weight. Reversing a link would mean something different. Define incoming volume as the sum of counts of links ending at a ward. Define transfers from active wards as the sum of counts on incoming links whose source had `active_cluster_at_start` = 1. These features use only information available at the observation point.

For generating the later target only, standardise each of the four arrays below over **all 120 wards**, using its population mean and standard deviation, after the displayed rounding of source columns. The data generator uses the following simulated log-odds for ward i :

$$-1.12 + 0.62z(\text{occupancy_ratio}_i) + 0.54z(\text{device_use_share}_i) + 0.24z(\text{incoming_volume}_i) + 0.86z(\text{transfers_from_active_wards}_i)$$

Convert log-odds to probability with the logistic function and independently draw a Bernoulli outcome for each ward without an active cluster. Assign blank outcomes to the other wards. This is the **construction** rule, not a model or scaling procedure for students to replicate. Student models must learn their own scaling using training observations only. A high volume of incoming transfers and transfers from initially affected wards may correlate; neither was derived from future labels.

Interpretation limits

The arbitrary probability rule makes network information potentially predictive in this particular fixed sample. Changes in held-out scores may be small or mixed. The file is too small and idealised for a healthcare decision, and a score change cannot show that transfers cause infections. There are no patient records, contact tracing, pathogen dynamics, infection-control interventions or independent validation hospital.